

NIM : 245410034

Nama : Irvan Cahya Nugraha

Kelas : IF-1

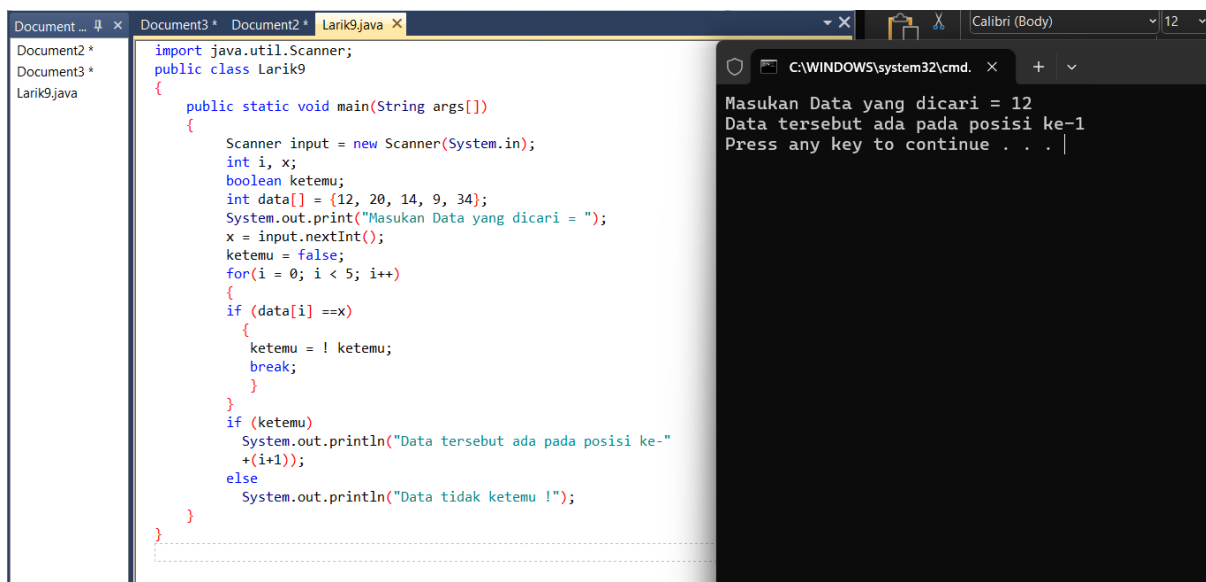
LISTING PRAKTIKUM ALGORITMA DAN PEMROGRAMAN LANJUT

MODUL 14

SEARCHING

PRAKTIK

Praktik 1



The screenshot shows a Java IDE with a file named `Larik9.java` open. The code implements a linear search algorithm. It imports `java.util.Scanner`, defines a `main` method, and creates a `Scanner` object. An array `data` is initialized with the values `{12, 20, 14, 9, 34}`. The program prompts the user to enter a target value `x`. It then iterates through the array, checking if any element matches `x`. If a match is found, it prints the position (index + 1) and sets a `ketemu` (found) flag to `true`. If no match is found after the loop, it prints a message indicating the data was not found. To the right of the IDE, a command prompt window shows the output of the program: `Masukan Data yang dicari = 12`, `Data tersebut ada pada posisi ke-1`, and `Press any key to continue . . .`.

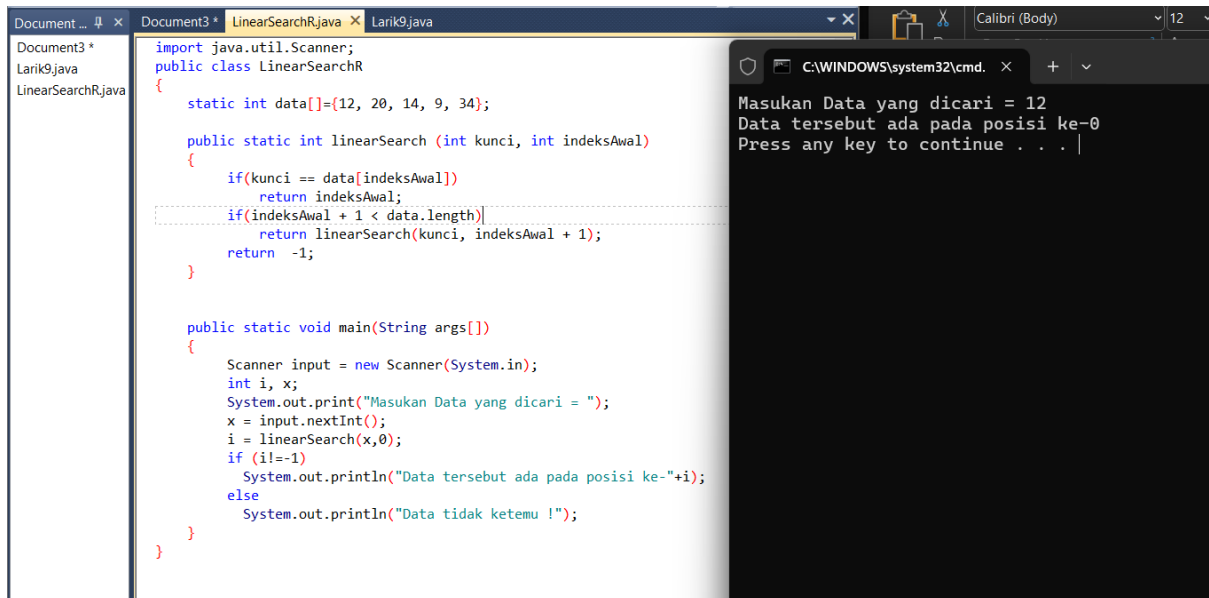
```
import java.util.Scanner;
public class Larik9
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        int i, x;
        boolean ketemu;
        int data[] = {12, 20, 14, 9, 34};
        System.out.print("Masukan Data yang dicari = ");
        x = input.nextInt();
        ketemu = false;
        for(i = 0; i < 5; i++)
        {
            if (data[i] ==x)
            {
                ketemu = ! ketemu;
                break;
            }
        }
        if (ketemu)
            System.out.println("Data tersebut ada pada posisi ke-"
                               +(i+1));
        else
            System.out.println("Data tidak ketemu !");
    }
}
```

C:\WINDOWS\system32\cmd. x + v
Masukan Data yang dicari = 12
Data tersebut ada pada posisi ke-1
Press any key to continue . . . |

Praktik 2

LATIHAN

Latihan 1



The screenshot shows an IDE with two tabs: 'LinearSearchR.java' and 'Larik9.java'. The 'LinearSearchR.java' tab is active, displaying the following Java code:

```
import java.util.Scanner;
public class LinearSearchR
{
    static int data[]={12, 20, 14, 9, 34};

    public static int linearSearch (int kunci, int indeksAwal)
    {
        if(kunci == data[indeksAwal])
            return indeksAwal;
        if(indeksAwal + 1 < data.length)
            return linearSearch(kunci, indeksAwal + 1);
        return -1;
    }

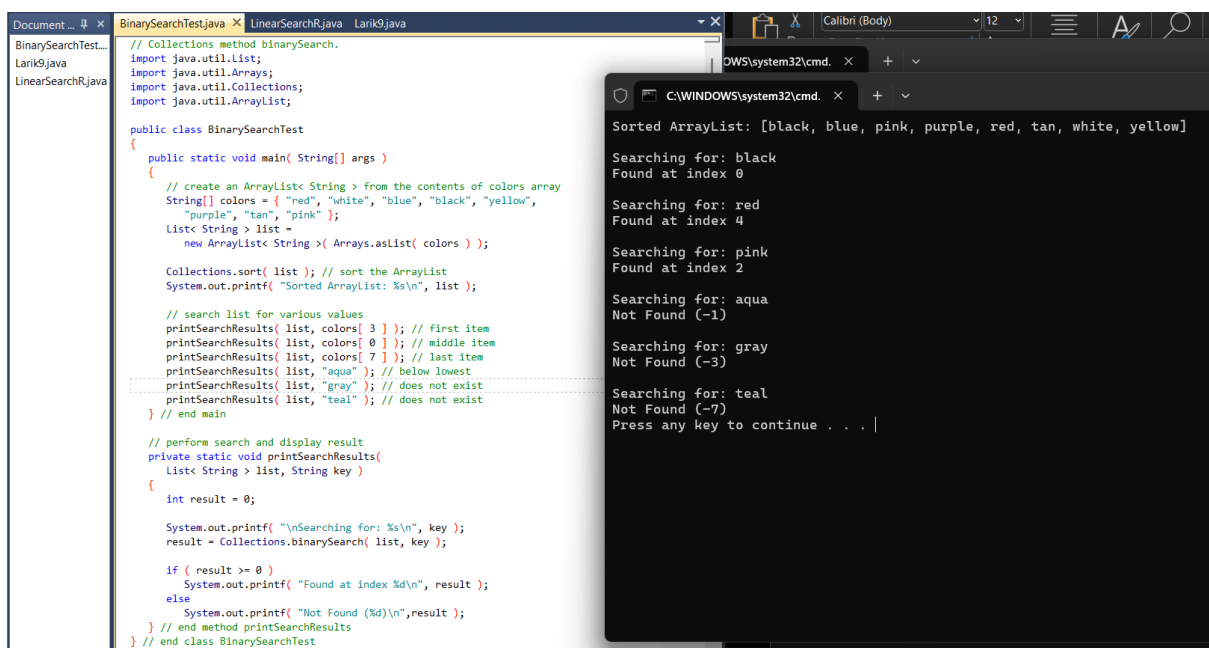
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        int i, x;
        System.out.print("Masukan Data yang dicari = ");
        x = input.nextInt();
        i = linearSearch(x,0);
        if (i!=-1)
            System.out.println("Data tersebut ada pada posisi ke-"+i);
        else
            System.out.println("Data tidak ketemu !");
    }
}
```

The 'Larik9.java' tab is also visible, showing the same code. The output window on the right shows the execution results:

```
Masukan Data yang dicari = 12
Data tersebut ada pada posisi ke-0
Press any key to continue . . . |
```

TUGAS

Tugas 1



The screenshot shows an IDE with two tabs: 'BinarySearchTest.java' and 'LinearSearchR.java'. The 'BinarySearchTest.java' tab is active, displaying the following Java code:

```
// Collections method binarySearch.
import java.util.List;
import java.util.Arrays;
import java.util.Collections;
import java.util.ArrayList;

public class BinarySearchTest
{
    public static void main( String[] args )
    {
        // create an ArrayList< String > from the contents of colors array
        String[] colors = { "red", "white", "blue", "black", "yellow",
            "purple", "tan", "pink" };
        List< String > list =
            new ArrayList< String >( Arrays.asList( colors ) );

        Collections.sort( list ); // sort the ArrayList
        System.out.printf( "Sorted ArrayList: %s\n", list );

        // search list for various values
        printSearchResults( list, colors[ 3 ] ); // first item
        printSearchResults( list, colors[ 0 ] ); // middle item
        printSearchResults( list, colors[ 7 ] ); // last item
        printSearchResults( list, "aqua" ); // below lowest
        printSearchResults( list, "gray" ); // does not exist
        printSearchResults( list, "teal" ); // does not exist
    } // end main

    // perform search and display result
    private static void printSearchResults(
        List< String > list, String key )
    {
        int result = 0;

        System.out.printf( "\nSearching for: %s\n", key );
        result = Collections.binarySearch( list, key );

        if ( result >= 0 )
            System.out.printf( "Found at index %d\n", result );
        else
            System.out.printf( "Not Found (%d)\n", result );
    } // end method printSearchResults
} // end class BinarySearchTest
```

The 'LinearSearchR.java' tab is also visible, showing the same code. The output window on the right shows the execution results:

```
Sorted ArrayList: [black, blue, pink, purple, red, tan, white, yellow]

Searching for: black
Found at index 0

Searching for: red
Found at index 4

Searching for: pink
Found at index 2

Searching for: aqua
Not Found (-1)

Searching for: gray
Not Found (-3)

Searching for: teal
Not Found (-7)
Press any key to continue . . . |
```

